Postcoital cervical mucus penetration test

There is insufficient evidence supporting the predictive value or clinical utility of this test. More studies are needed to support improved outcomes (i.e., increased successful pregnancies with delivery of liveborn children).

The postcoital test, also known as the cervical mucous penetration assay, is designed to evaluate impaired male factors such as hyperviscosity, abnormal penile anatomy, decreased semen volume despite good sperm density, and unexplained infertility. The postcoital test is performed by examining the cervical mucus several hours after intercourse, ideally around the periovulatory phase of the female. The cervical mucus is smeared on a slide and examined microscopically for the presence of ferning within the mucus and for the quantitative presence of sperm and forward sperm motility. The role of the postcoital test continues to be a point of debate (Mohit, K. 2006).

Reactive oxygen species (ROS) testing

There is insufficient evidence supporting the predictive value or clinical utility of this test. Additional studies are needed to support improved clinical outcomes.

Reactive oxygen species have a significant impact on spermatogenesis as well as on sperm function. Supraphysiologic levels of ROS can affect all aspects of the semen analysis. Multiple studies have demonstrated the detrimental effects of ROS on sperm concentration, motility, morphology and ROS are implicated in sperm DNA damage and apoptosis. Conversely, other studies have demonstrated no correlation between ROS and sperm motility. Recent studies have demonstrated that ROS values have a sensitivity of 68.8% and specificity of 93.8% in diagnosing male factor infertility (infertility due to abnormal semen parameters, not female-related factors) (Ko, E. et. al. 2014).

Sperm Acrosome reaction test

There is insufficient evidence supporting the predictive value or clinical utility of this test. More studies are needed to support improved outcomes (i.e., increased successful pregnancies with delivery of liveborn children).

The acrosome reaction is induced by spermatozoa-ZP binding, right after the lytic acrosomal enzymes are released, and the spermatozoa proceed to the zonal matrix with increased flagellar motility. Sperm-ZP binding tests including sperm-zona binding test and hemizona test (HZA) and acrosome reaction tests have been shown to provide clinically significant information to predict IVF results. The highest specificity and sensitivity associated with sperm-ovum interaction are also provided by the sperm-ZP binding tests. Although there is a significant relation between the in vitro results of these tests and fertilization, there are problems about their utilization in routine clinical practice such as difficulties in terms of requirements for human material requirement, challenging occasionally time-consuming and expensive techniques (Fuat, K. & Baris, A. 2017).